# CFO Payroll Intelligence Dashboard

## Executive Summary – Week 1 Deliverable

### ◇ Business Objective

The objective of this project was to design and develop a CFO-level Payroll Intelligence Dashboard to provide executive visibility into payroll expenses, department-wise cost distribution, employee cost metrics, and workforce efficiency indicators. The dashboard enables leadership to monitor total payroll expenditure, identify high-cost departments, evaluate overtime impact, and support data-driven financial planning decisions.

### ◇ Data Sources

The dashboard was built using the following core datasets:

• Employees Table – Contains employee details such as employee_id, department, designation, and joining_date.

• Payroll Table – Includes salary components such as gross_salary, net_salary, tax_deduction, pf_deduction, and payroll metrics.

• Attendance Table – Stores attendance-related data including working_days, present_days, and overtime_hours.

### ◇ KPI Definitions (Business Meaning)

• Total Payroll Cost – Total net salary expenditure incurred by the company.

• Total Gross Payroll – Total salary before deductions.

• Cost Per Employee – Average payroll cost per employee.

• Department Expense – Total payroll cost allocated to each department.

• Department Cost Per Employee – Average employee cost within each department.

• Overtime Percentage – Proportion of overtime relative to total working capacity.

• Absenteeism Rate – Percentage of working days lost due to employee absence.

• Headcount by Department – Total number of employees within each department.

• Total Tax Deduction – Total tax liability deducted from payroll.

• Total PF Deduction – Total provident fund contribution deducted from payroll.

### ◇ Key Insights

• IT department contributes approximately 38% of the total payroll, making it the highest cost center.

• Overtime contributes around 20% to workforce capacity, indicating potential cost optimization opportunities.

• Average cost per employee is approximately ₹56.15K.

• Payroll cost distribution shows moderate concentration in technical departments.

• Attendance metrics highlight absenteeism trends that may impact productivity and financial planning.

### ◇ Optimization Strategy

To ensure efficient real-time dashboard performance, the following optimization strategies were implemented:

• Indexed employee_id in core tables to improve join performance.

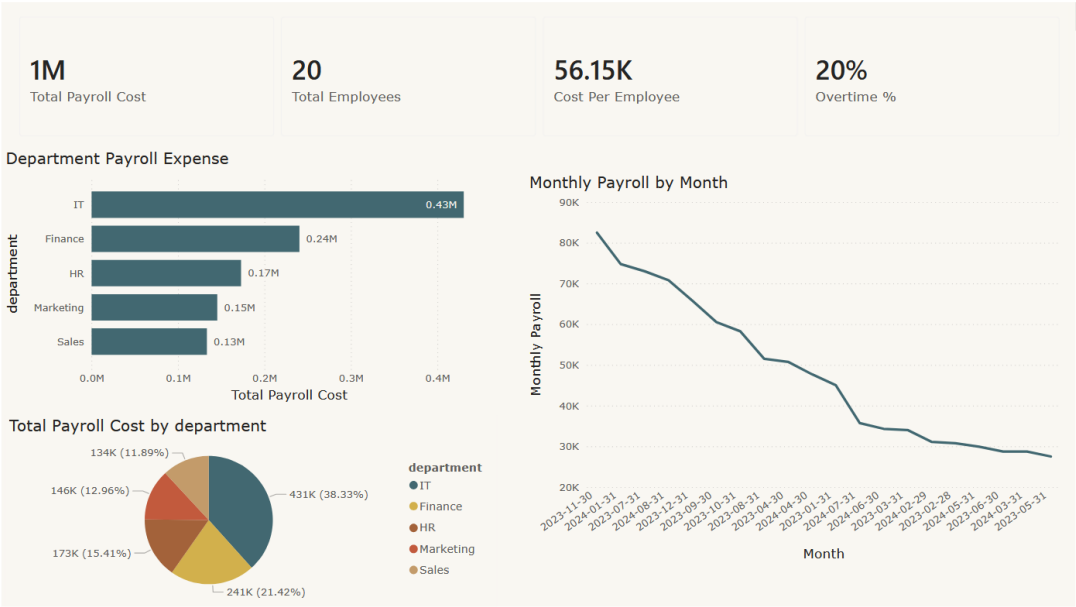
• Created aggregated SQL views (vw_cfo_summary, vw_department_expense, vw_attendance_summary) to reduce heavy runtime computations.

• Avoided SELECT * and retrieved only required columns to minimize data load.

• Reduced join complexity by pre-aggregating department and attendance metrics.

• Used summarized measures instead of row-level calculations within the dashboard tool.

## Dashboard Overview:




**Conclusion:** The CFO Payroll Intelligence Dashboard delivers executive-ready insights by combining optimized SQL queries, structured KPI definitions, and performance-focused data modeling. The solution demonstrates both technical proficiency and strong business understanding aligned with financial decision-making.